

Seperable equations

a 1st order seperable equation of the form $\frac{dy}{dx} = g(x)h(y)$ is said to be separable or have separable variables

Step one: get all the x and y on one side including dy and dx

$$\frac{dy}{dx} = g(x)h(y) \Rightarrow \frac{1}{h(y)} dy = g(x) dx$$

all y 's are
on one side
including dy

all the x 's are
on one side including
the dx

Step two: take the integral on both sides

$$\frac{1}{h(y)} dy = g(x) dx \Rightarrow \int \frac{1}{h(y)} dy = \int g(x) dx$$

$$\Rightarrow \ln|h(y)| = G(x) + C$$

to know which letter we are solving for we look at the numerator of $\frac{dy}{dx}$
here we are solving for " y " so we put the "+ C" on the other
letter

Step three: solve for " y "

$$\ln|h(y)| = G(x) + C \Rightarrow e^{\ln|h(x)|} = e^{G(x)+C}$$

$$e^{\ln(x)} = x$$

$$\Rightarrow |h(y)| = e^{G(x)} \cdot e^C \quad \text{let } e^C = C_1 \Rightarrow \boxed{h(y) = \pm C_1 e^{G(x)}}$$

a absolute value measures the distance from zero so if we
wanted to measure $|-3|$ we get 3 because thats how far its from
zero but if we go three digits from zero on both side we get ± 3

Practice Problems

$$1) \quad (x^2+1) \frac{dy}{dx} = xy$$

$$\frac{1}{y} dy = \frac{x}{(x^2+1)} dx \Rightarrow \int \frac{1}{y} dy = \int \frac{x}{(x^2+1)} dx \Rightarrow \begin{aligned} u &= x^2+1 \\ du &= 2x dx \\ \frac{du}{2} &= dx \end{aligned}$$

$$\Rightarrow \ln|y| = \frac{1}{2} \int \frac{\cancel{x}}{u} \cdot \frac{du}{\cancel{x}} \Rightarrow \ln|y| = \frac{1}{2} \ln|x^2+1| + C$$

$$\Rightarrow e^{\ln|y|} = e^{\ln|x^2+1|^{1/2} + C} \Rightarrow \cancel{e^{\ln|y|}} = \cancel{e^{\ln|x^2+1|}} \cdot e^C$$

$$\Rightarrow \text{let } e^C = C_1 \quad |y| = C_1 \sqrt{x^2+1} \Rightarrow \boxed{y = \pm C_1 \sqrt{x^2+1}}$$

$$2) \quad \frac{dy}{dx} = ye^x, \quad y(0) = 2e$$

Find the general solution and the IVP

• this is a general solution they did not say solve for anything

$$\frac{1}{y} dy = e^x dx \Rightarrow \int \frac{1}{y} dy = \int e^x dx \Rightarrow \ln|y| = e^x + C$$

$$y(0) = 2e$$

x y

$$\ln|2e| = e^0 + C \Rightarrow \ln|2| + \ln|e| = C \Rightarrow C = \ln(2)$$

$$\ln|y| = e^x + C \Rightarrow \boxed{\ln|y| = e^x + \ln(2)}$$

if you are going to replace e^C you need to write the statement:

$$\text{let } e^C = C_1$$

When you are told to solve for a general solution that means finding a formula that represents all possible solution

$$3) \quad \frac{dy}{dx} = \frac{xy + 3x - y - 3}{xy - 2x + 4y - 8} \quad \text{Find the general solution}$$

$$\frac{dy}{dx} = \frac{x(y+3) - (y+3)}{x(y-2) + 4(y-2)} \Rightarrow \frac{dy}{dx} = \frac{(x-1)(y+3)}{(x+4)(y-2)} \Rightarrow \frac{(y-2)}{(y+3)} dy = \frac{(x-1)}{(x+4)} dx$$

$$\Rightarrow \int \frac{(y-2)}{(y+3)} dy = \int \frac{(x-1)}{(x+4)} dx$$

You need to use long division to solve this you really cannot do it any other way

$$\begin{array}{r} 1 \leftarrow Q \\ y+3 \overline{) y-2} \\ - (y+3) \\ \hline -7 \leftarrow R \end{array}$$

$$\left(Q + \frac{R}{d} \right)$$

$$\left(1 - \frac{5}{y+3} \right)$$

$$\begin{array}{r} 1 \leftarrow Q \\ x+4 \overline{) x-1} \\ - (x+4) \\ \hline -5 \leftarrow R \end{array}$$

$$\left(Q + \frac{R}{d} \right)$$

$$\left(1 - \frac{5}{x+4} \right)$$

$$\int \left(1 - \frac{5}{y+3} \right) dy = \int \left(1 - \frac{5}{x+4} \right) dx \Rightarrow y - 5 \ln|y+3| = x - 5 \ln|x+4| + C$$

$$y - x = 5 \ln|y+3| - 5 \ln|x+4| + C \Rightarrow y - x = \ln \left[\frac{(y+3)^5}{(x+4)^5} \right] + C$$

$$e^{y-x} = e^{\ln \left[\frac{(y+3)^5}{(x+4)^5} \right] + C} \Rightarrow e^{y-x} = e^{\ln \left[\frac{(y+3)^5}{(x+4)^5} \right]} \cdot e^C$$

let $e^C = C_1$

$$e^{y-x} = C_1 \frac{(y+3)^5}{(x+4)^5}$$